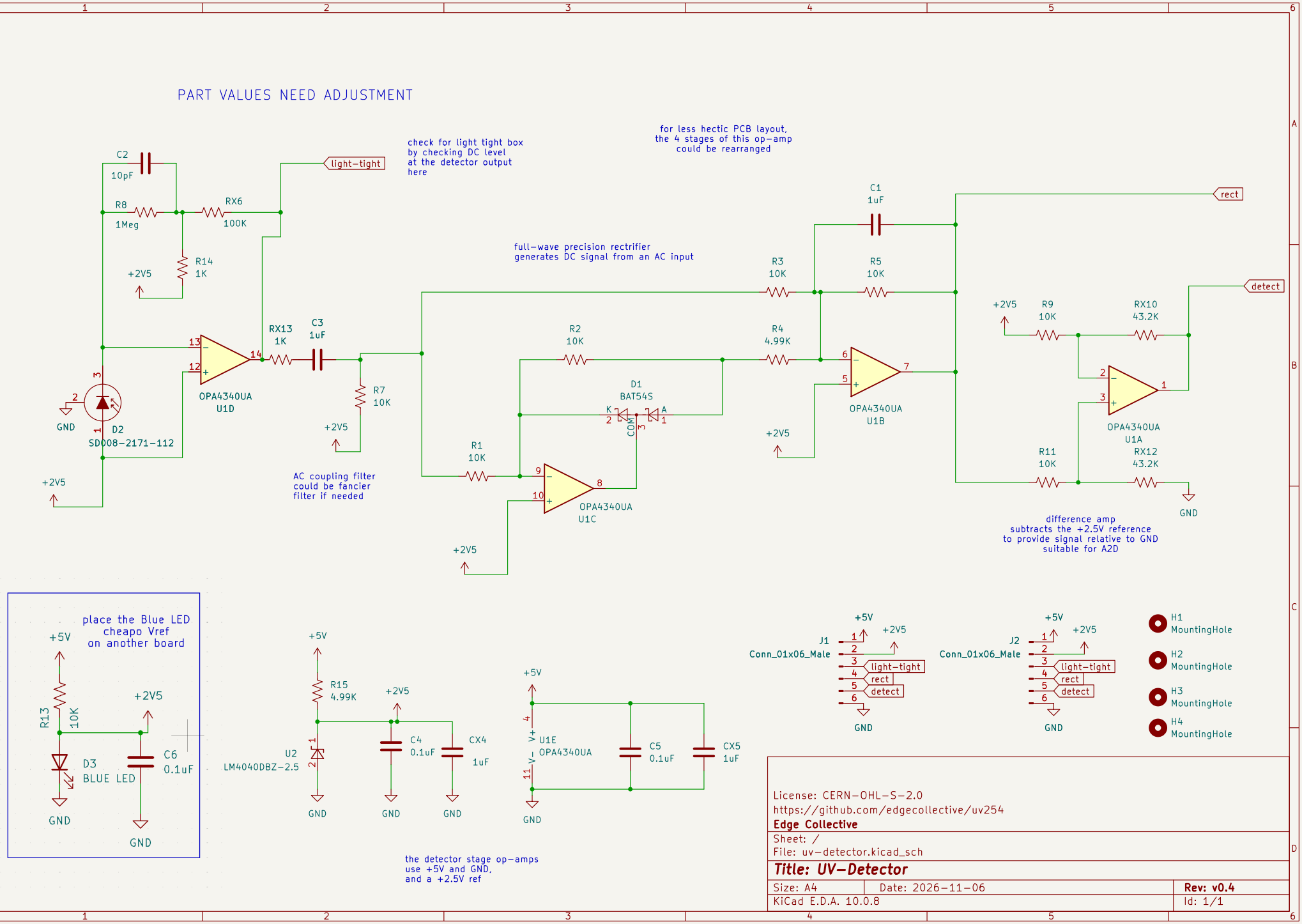

PART VALUES NEED ADJUSTMENT
check for light tight box
by checking DC level
at the detector output
here
for less hectic PCB layout,
the 4 stages of this op-amp
could be rearranged
full-wave precision rectrifier
generates DC signal from an AC input
C2
10pF
R8
1Meg
RX6
100K
light-tight
R14
1K
+2V5
RX13
1K
C3
1uF
R7
10K
OPA4340UA
U1D
D2
SD008-2171-112
GND
AC coupling filter
could be fancier
filter if needed
R2
10K
R1
10K
D1
BAT54S
OPA4340UA
U1C
C1
1uF
R3
10K
R5
10K
R4
4.99K
OPA4340UA
U1B
rect
detect
R9
10K
RX10
43.2K
OPA4340UA
U1A
R11
10K
RX12
43.2K
GND
difference amp
subtracts the +2.5V reference
to provide signal relative to GND
suitable for A2D
place the Blue LED
cheapo Vref
on another board
R13
10K
D3
BLUE LED
C6
0.1uF
R15
4.99K
U2
LM4040DBZ-2.5
C4
0.1uF
CX4
1uF
U1E
OPA4340UA
C5
0.1uF
CX5
1uF
J1
Conn_01x06_Male
J2
Conn_01x06_Male
light-tight
rect
detect
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
the detector stage op-amps
use +5V and GND,
and a +2.5V ref
License: CERN-OHL-S-2.0
https://github.com/edgecollective/uv254
Edge Collective
Sheet: /
File: uv-detector.kicad_sch
Title: UV-Detector
Size: A4
Date: 2026-11-06
Rev: v0.4
KiCad E.D.A. 10.0.8
Id: 1/1